



Innovations in Industrial Fluid Sealing

ULTRA NE 1005

Advanced Ultra-Low Emission All-Braided Packing for High-Performance Industrial Sealing Applications

ULTRA NE 1005

Inmarco ULTRA NE 1005 is an advanced ultra-low emission all-braided compression packing specifically engineered for high-performance industrial sealing applications requiring exceptional fugitive emission control, superior chemical resistance, and long-term operational reliability. Manufactured from premium expanded flexible graphite with integrated Inconel wire mesh reinforcement and advanced lubrication technology, ULTRA NE 1005 delivers unmatched sealing performance under extreme pressure, temperature, and chemical exposure conditions.



The packing's sophisticated construction combines high-purity expanded graphite fibers with corrosion-resistant Inconel wire mesh reinforcement, providing outstanding mechanical strength, dimensional stability, and adaptability to shaft movement and thermal cycling. Its advanced lubrication system ensures smooth operation, minimal friction, reduced shaft wear, and extended service life across a wide range of dynamic and static sealing applications.

ULTRA NE 1005 is tested and certified as per API 622 3rd edition and ISO 15848 with CC1 endurance class and tightness class AH, ensuring compliance with the most stringent international fugitive emission standards. The packing is fire-safe compliant and suitable for critical service applications in petrochemical, refinery, power generation, chemical processing, and other demanding industrial environments where zero-emission performance and operational safety are paramount.

Testing & Certifications

Standard / Certification	Performance Achievement
API 622 3rd edition	Certified for ultra-low emission valve packing applications in critical service
ISO 15848 – CC1 Endurance Class, Tightness Class AH	Tested and certified for fugitive emission performance with highest tightness classification
Fire Safety Compliance	Fire-safe compliant for critical high-temperature and hydrocarbon applications

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Technical Properties

Property	Specification
Construction	All-braided expanded graphite with Inconel wire mesh reinforcement
Material Composition	High-purity expanded flexible graphite + Inconel wire mesh
pH Range	0 – 14
Temperature Range – Oxidizing Media	-196°C to +538°C
Temperature Range – Steam Service	Up to +650°C
Maximum Pressure	Up to 500 bar
Size Range	3 mm² to 25 mm²
Chemical Compatibility	Wide Chemical Compatibility, for details contact Product Engineering
Fire Safety	Fire-safe compliant

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Key Features

- Advanced all-braided construction with high-purity expanded graphite and Inconel wire mesh reinforcement
- Certified as per API 622 3rd edition and ISO 15848 CC1 endurance class with tightness class AH
- Exceptional fugitive emission control performance suitable for critical service applications
- Superior thermal stability from -196°C to +650°C depending on service media
- Fire-safe compliant for high-temperature hydrocarbon and steam service applications
- Low friction characteristics reduce shaft and equipment wear during continuous operation
- Excellent chemical resistance across a broad pH range and aggressive industrial media
- Inconel wire mesh reinforcement provides superior mechanical strength and pressure resistance
- Dimensionally stable construction maintains sealing integrity under thermal cycling and mechanical stress
- Advanced lubrication technology ensures smooth start-up and extended operational life
- Suitable for rotating, reciprocating, and static sealing applications in high-pressure environments

Applications

- Critical service valves in petrochemical and refinery operations
- High-pressure pumps handling hydrocarbons, chemicals, and aggressive process fluids
- Steam turbines and power generation equipment
- Chemical processing reactors and vessels
- Fugitive emission-controlled equipment requiring API 622 and ISO 15848 compliance
- High-temperature and high-pressure rotating and reciprocating equipment
- Oil and gas production and processing facilities
- Industrial boilers and heat exchangers
- Compressors and expanders in gas processing applications
- Environmental compliance-driven sealing applications in refineries and chemical plants
- Fire-safe service applications requiring reliable sealing performance under emergency conditions

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Performance Benefits

- ✓ Delivers exceptional fugitive emission control performance certified to API 622 and ISO 15848 standards
- ✓ Maintains reliable sealing integrity across a wide temperature range from cryogenic to extreme high-temperature conditions
- ✓ Inconel wire mesh reinforcement provides superior mechanical strength, dimensional stability, and pressure resistance
- ✓ Low friction characteristics minimize shaft and equipment wear, reducing maintenance costs and downtime
- ✓ Advanced lubrication technology ensures smooth operation and extended service life
- ✓ Fire-safe compliance provides critical safety protection in high-temperature hydrocarbon and steam service applications
- ✓ Excellent adaptability to shaft movement, thermal cycling, and pressure fluctuations
- ✓ Suitable for critical service applications requiring the highest levels of fugitive emission control and operational reliability
- ✓ Reduces environmental impact by minimizing leakage and emissions from industrial equipment
- ✓ Provides consistent sealing performance under demanding process conditions in petrochemical, refinery, and chemical industries
- ✓ Supports regulatory compliance with international fugitive emission standards and environmental regulations
- ✓ Extends equipment life and operational efficiency through superior sealing performance and reduced maintenance requirements

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All technical information, recommendations, and performance data provided in this technical datasheet are based on current knowledge, laboratory testing, and/or industrial experience. Actual product performance may vary depending on operating conditions, equipment design, installation practices, process parameters. Final product selection and suitability should be determined by the user based on the specific service conditions and application requirements.